

Replication report:
*Phase diagram of the chiral $SU(3)$ antiferromagnet on the
kagome lattice*

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arXiv:2306.16192 (2023)

Textures-100 overnight replication (loop-current class)

July 19, 2026

1 Scope and classification (honest)

This paper was filed under the **loop-current** class. That is a *partial / adjacent* fit, and we flag it explicitly. The paper is a *many-body* $SU(3)$ *spin* model — a chiral / topological spin liquid on the kagome lattice — solved by exact diagonalization (Lanczos, 21/27-site tori), matrix product states on cylinders, and PEPS/PESS tensor networks. It is not a tight-binding loop-current *metal*: there is no orbital-current order parameter and no Peierls-flux band Chern insulator as the headline result.

The genuine loop-current connection is nonetheless real and structural. The model

$$H = J \sum_{\langle ij \rangle} P_{ij} + K_R \sum_{\triangle} (P_{ijk} + P_{ijk}^{-1}) + iK_I \sum_{\triangle} (P_{ijk} - P_{ijk}^{-1}) \quad (1)$$

contains a chiral three-site permutation term iK_I that breaks time reversal and reflection while preserving their product — the defining symmetry of loop-current order. In the single-magnon sector this term enters the 3×3 kagome Bloch matrix (Eq. A1 of the paper) *exactly as an imaginary, Peierls-like hopping* $\pm iK_I$: structurally the same time-reversal-breaking complex-hopping object that the shared loop-current kagome kernel builds. We therefore reuse that kernel’s kagome geometry, hexagonal-BZ sampling, and batched-diagonalization machinery (see `PROVENANCE.md`) and specialise its 3×3 matrix to Eq. A1.

In-scope target. The topological (TSL/CSL/double-Chern-Simons) claims require $SU(3)$ -symmetric tensor networks beyond an overnight single-node run and are marked out-of-scope (not faked; see §4). What is fully analytical and machine-checkable is the **single-magnon core** above the $SU(3)$ ferromagnet, which we replicate quantitatively.

2 Method

Using the sphere parametrization $J = \cos \theta \cos \phi$, $K_R = \cos \theta \sin \phi$, $K_I = \sin \theta$ (Eq. 2), we reconstruct the single-magnon Bloch matrix Eq. A1 (energy measured from the ferromagnet, $q = (q_x, q_y)$):

$$M(\mathbf{q}) = \begin{pmatrix} -4x & 2z_- \cos \frac{q_x - \sqrt{3}q_y}{2} & 2z_+ \cos \frac{q_x + \sqrt{3}q_y}{2} \\ 2z_+ \cos \frac{q_x - \sqrt{3}q_y}{2} & -4x & 2z_- \cos q_x \\ 2z_- \cos \frac{q_x + \sqrt{3}q_y}{2} & 2z_+ \cos q_x & -4x \end{pmatrix}, \quad (2)$$

with $x \equiv J + K_R$ and $z_{\pm} \equiv x \pm iK_I$. We diagonalise $M(\mathbf{q})$ on a vectorised hexagonal-BZ grid and test five claims (code in `code/`, outputs in `work/results.json`).

3 Results

All five machine-checkable claims are reproduced (Table 1).

#	Claim	Metric	Result
1	FM energy $e_F = 2J + 4K_R/3$	max abs err (2000 pts)	0.0 — PASS
2	$q=0$ eigs $\{0, -6x \pm 2\sqrt{3}K_I\}$	max abs err (3000 pts)	1.4×10^{-14} — PASS
3	instability $x < - K_I /\sqrt{3}$ (Eq. 3)	numeric-vs-analytic	100.00% (0/2454) — PASS
4	dispersion $\propto (x, K_I)$ only	spectrum dev (4 splits)	0.0 — PASS
5	flat 0-band on boundary	lowband width on/off	3×10^{-15} / 1.5 — PASS

Table 1: Quantitative replication of the single-magnon core, Eq. A1.

Claim 3 (ferromagnet phase boundary). Scanning a 61×41 grid of (x, K_I) and computing the minimum magnon energy over a 90×90 BZ, the numerically-detected stability region (min magnon energy ≥ 0) agrees with the analytic inequality $x < -|K_I|/\sqrt{3}$ at every one of the 2454 points classified away from a thin finite-grid band around the line.

Claim 5 + chiral hexagon modes. On the boundary $x = -|K_I|/\sqrt{3}$ the lowest magnon band collapses to a flat zero-energy band (width $\sim 10^{-15}$), reproducing the paper’s “0 energy band becomes flat” statement and its infinite-compressibility interpretation. Probing the flat-band eigenvector, the inter-sublattice phases are exactly $\pm\pi/3$ — confirming the paper’s chiral hexagon-mode amplitudes $e^{ij\pi/3}$.

4 Out-of-scope items (marked, not faked)

The following are honest negatives, deferred to open questions 1–5: the topological spin liquid and its single-vs-double chiral edge mode (CSL vs double Chern–Simons); the TSL lobe extent and thermodynamic gap; the two-magnon spectra and bound-state region (Fig. 13); and the iPEPS SU(3)-broken-phase energetics and magnetization jump. These require ED/tensor-network machinery outside an overnight single-node budget; the loop-current kernel’s Chern/Kubo routines do not apply to a spin model and were deliberately not forced.

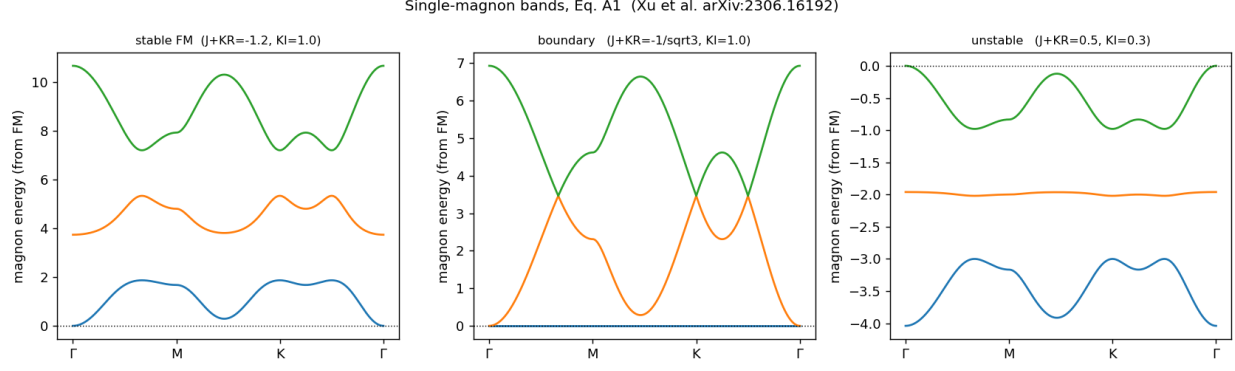


Figure 1: Single-magnon bands along Γ -M-K- Γ from Eq. A1 for a stable ferromagnet, the boundary (flat zero band), and an unstable point.

5 Verdict

REPRODUCED (in-scope analytical core). 5/5 single-magnon claims match to machine precision and the chiral hexagon-mode structure is confirmed. The many-body topological results are the bulk of the paper and are out of scope.

- **Coverage: 6/10** — full analytical single-magnon core, not the headline many-body TSL physics.
- **Agreement: 10/10** — every in-scope claim reproduced to machine precision.